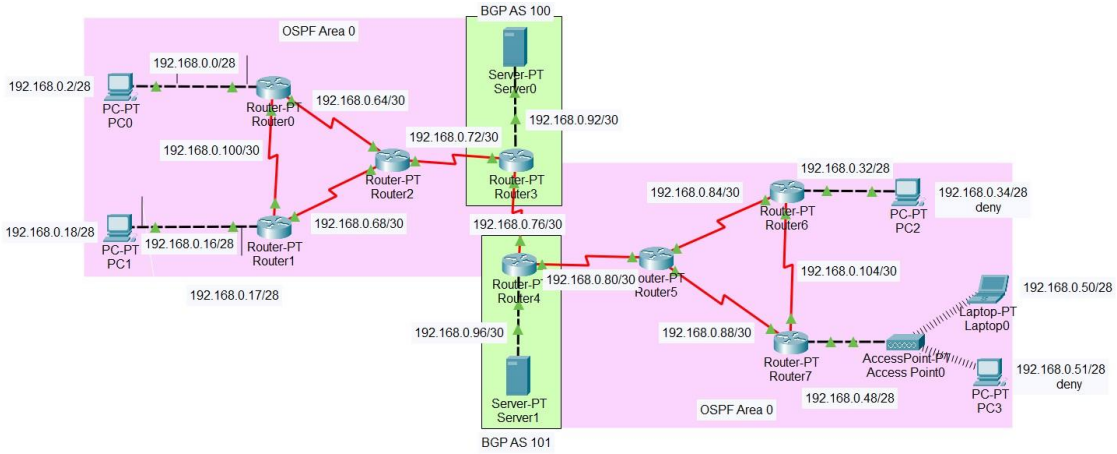
# BAB 1

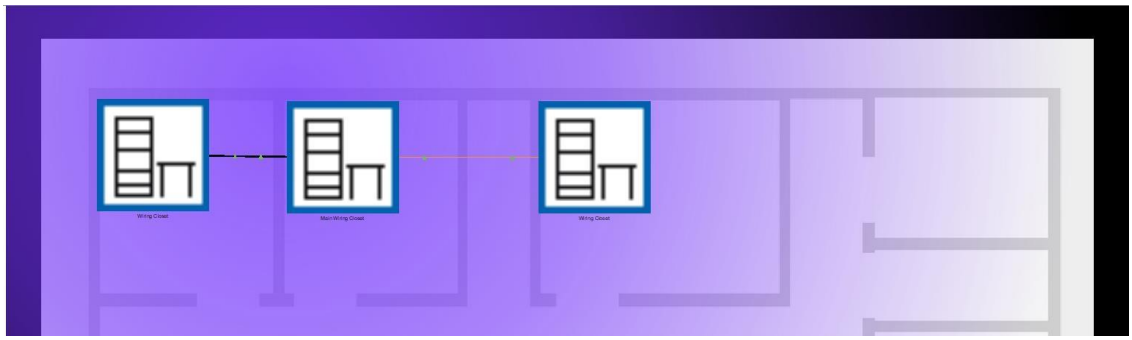
# Packet Tracer

## A. Topologi Jaringan

- **Logical**




- **Physical**



## B. Desain VLSM dan Implementasi (Static)

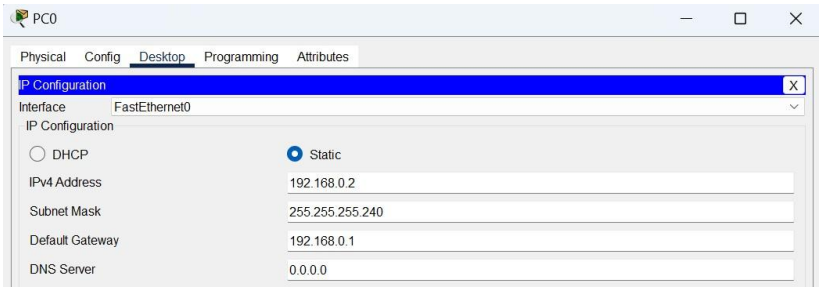
**192.168.0.0/24**

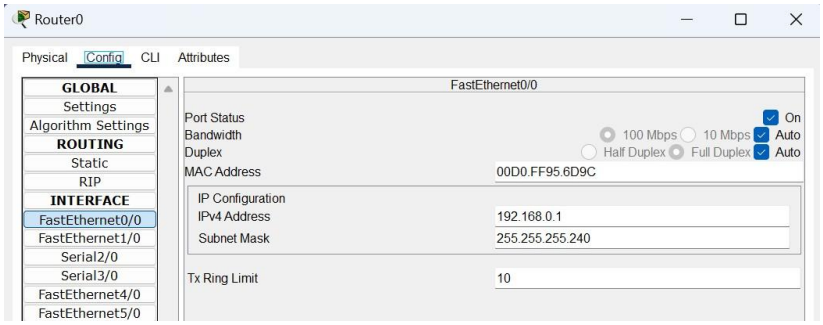
- **Subnet 1 (Router0 – PC0) /28**

  Subnet ID: 192.168.0.0

  Host: 192.168.0.1 – 192.168.0.14

  Broadcast: 192.168.0.15



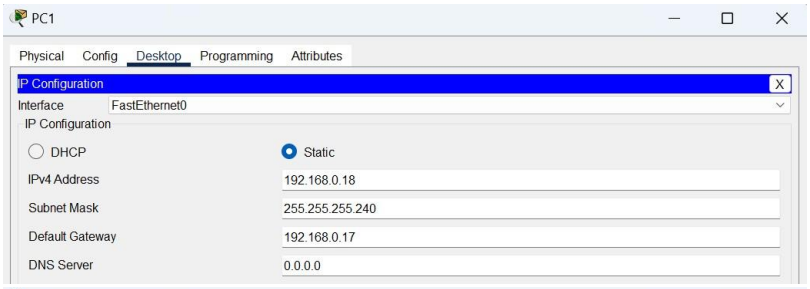


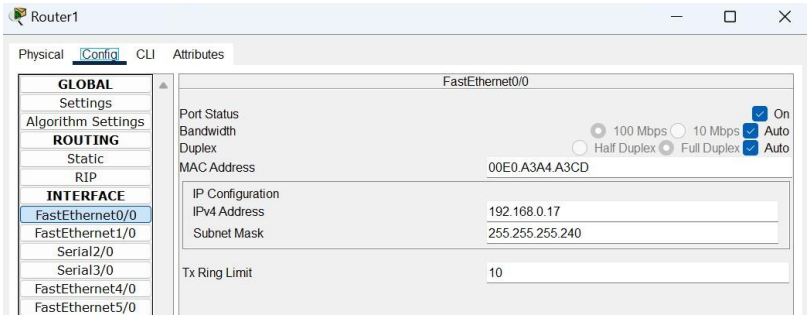

- **Subnet 2 (Router1 – PC1) /28**

  Subnet ID: 192.168.0.16

  Host: 192.168.0.17 – 192.168.0.30

  Broadcast: 192.168.0.31



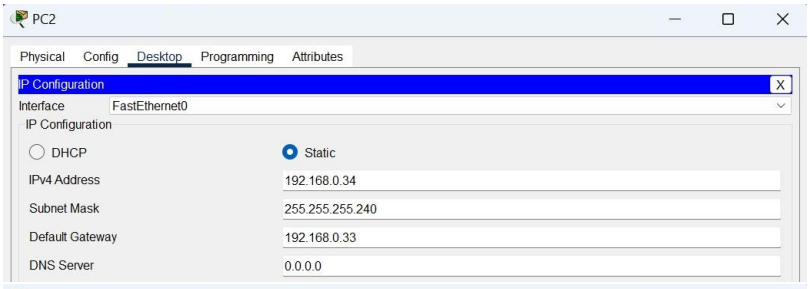



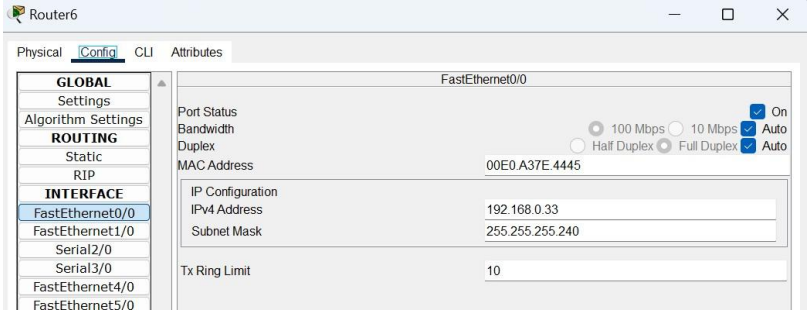

- **Subnet 3 (Router6 – PC2) /28**

  Subnet ID: 192.168.0.32

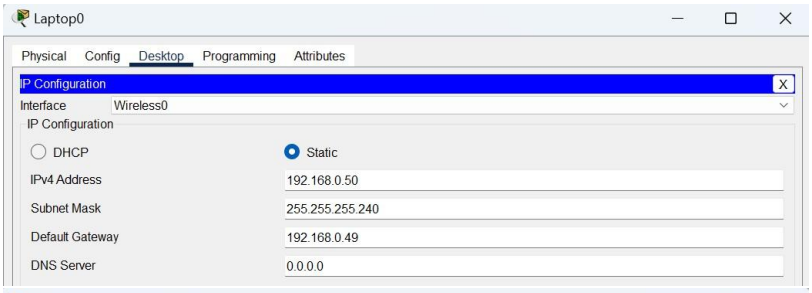
  Host: 192.168.0.33 – 192.168.0.46

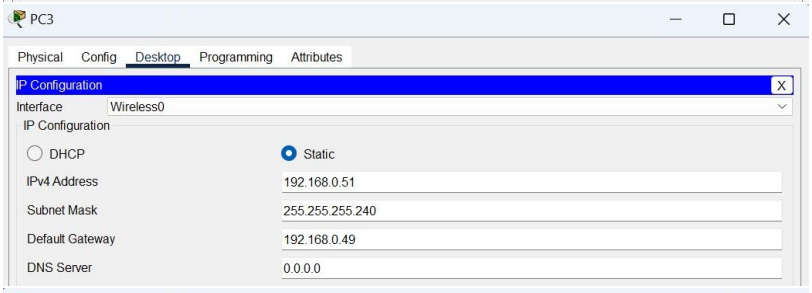
  Broadcast: 192.168.0.47

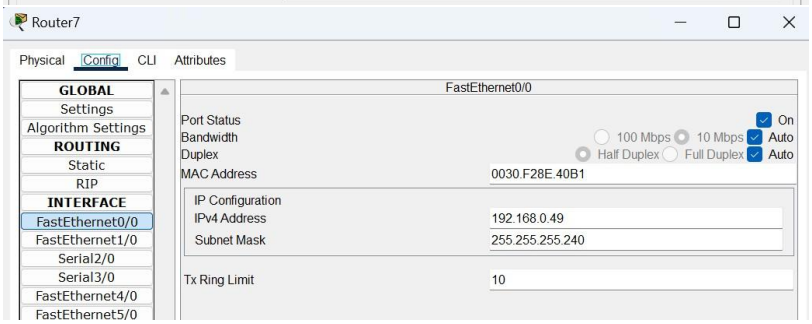


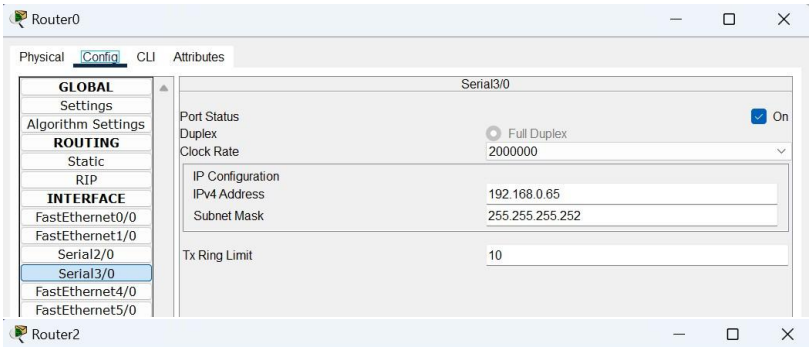
- **Subnet 4 (Router7 – Access Point) /28**
  Subnet ID: 192.168.0.48
  Host: 192.168.0.49 – 192.168.0.62
  Broadcast: 192.168.0.63

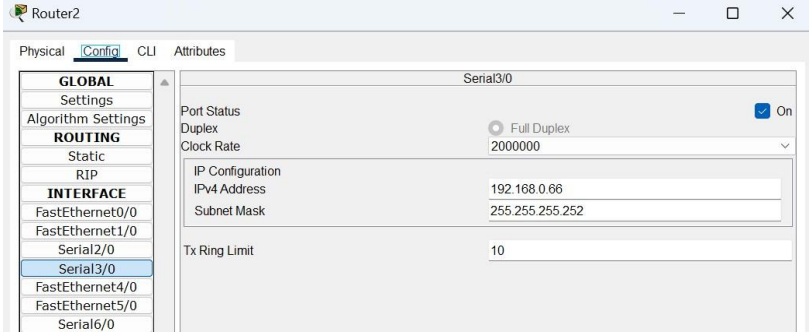


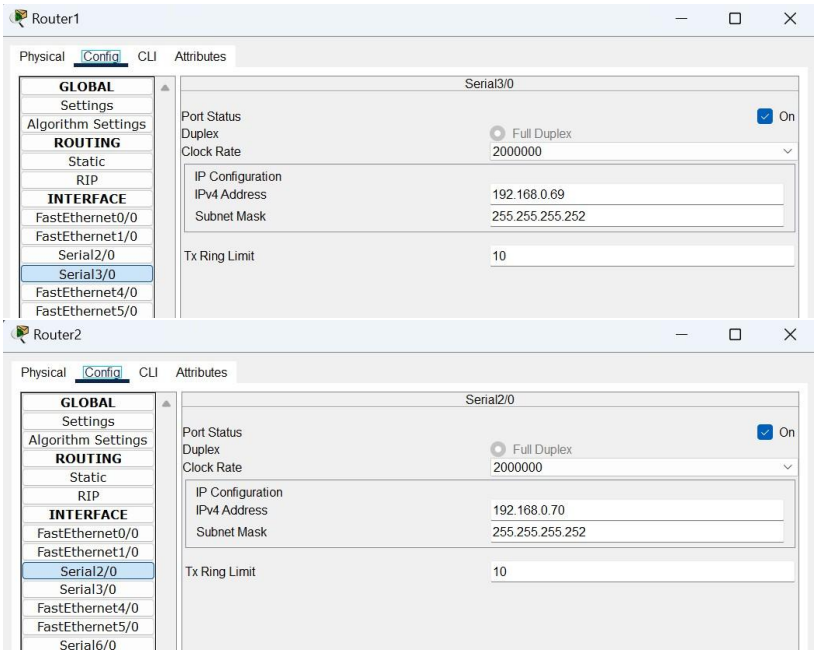




- **Subnet 5 (Router0 – Router2) /30**
  Subnet ID: 192.168.0.64
  Host: 192.168.0.65 – 192.168.0.66
  Broadcast: 192.168.0.67

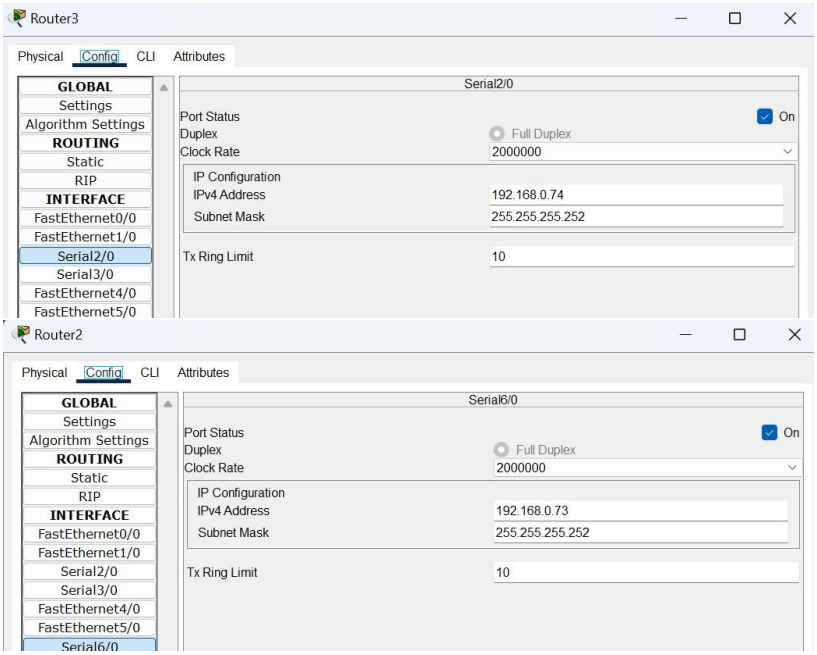


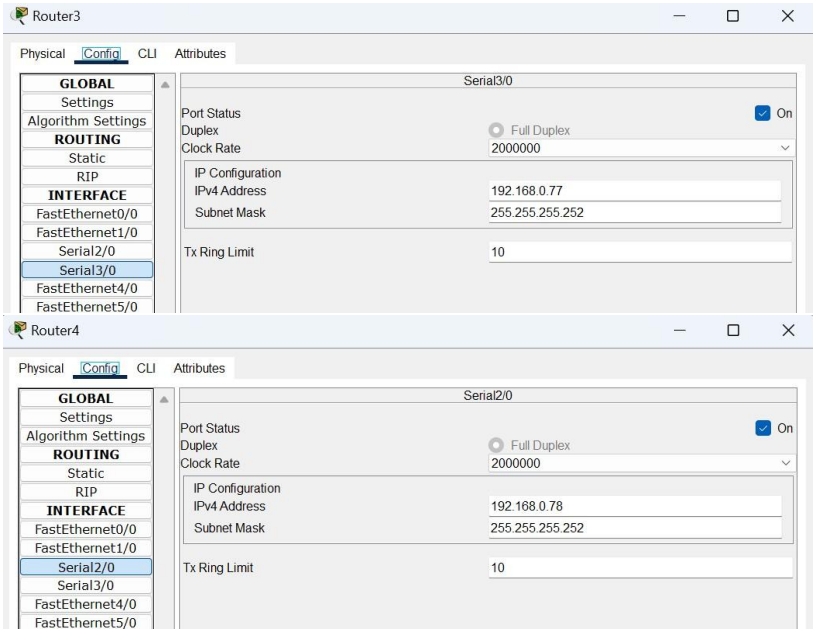
- **Subnet 6 (Router1 – Router2)  /30**
  Subnet ID: 192.168.0.68
  Host: 192.168.0.69 – 192.168.0.70
  Broadcast: 192.168.0.71




- **Subnet 7 (Router3 – Router2)  /30**
  Subnet ID: 192.168.0.72
  Host: 192.168.0.73 – 192.168.0.74
  Broadcast: 192.168.0.75




- **Subnet 8 (Router4 – Router3)  /30**
  Subnet ID: 192.168.0.76
  Host: 192.168.0.77 – 192.168.0.78
  Broadcast: 192.168.0.79

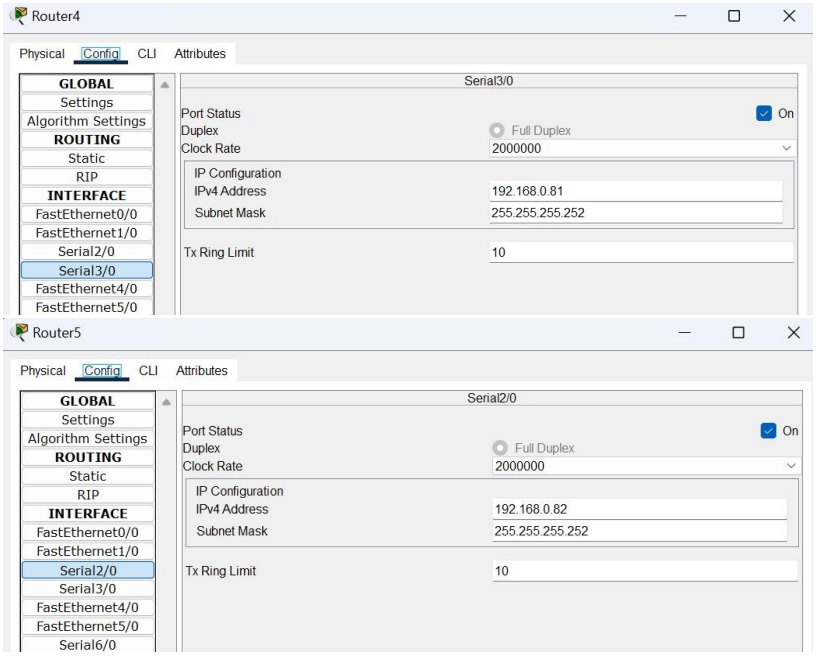



- **Subnet 9 (Router4 – Router5) /30**

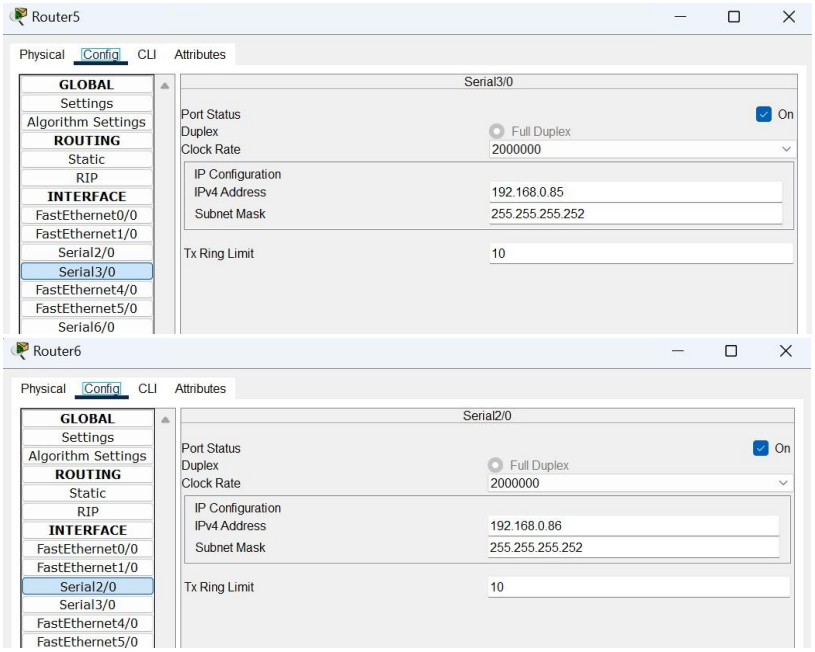
  Subnet ID: 192.168.0.80

  Host: 192.168.0.81 – 192.168.0.82

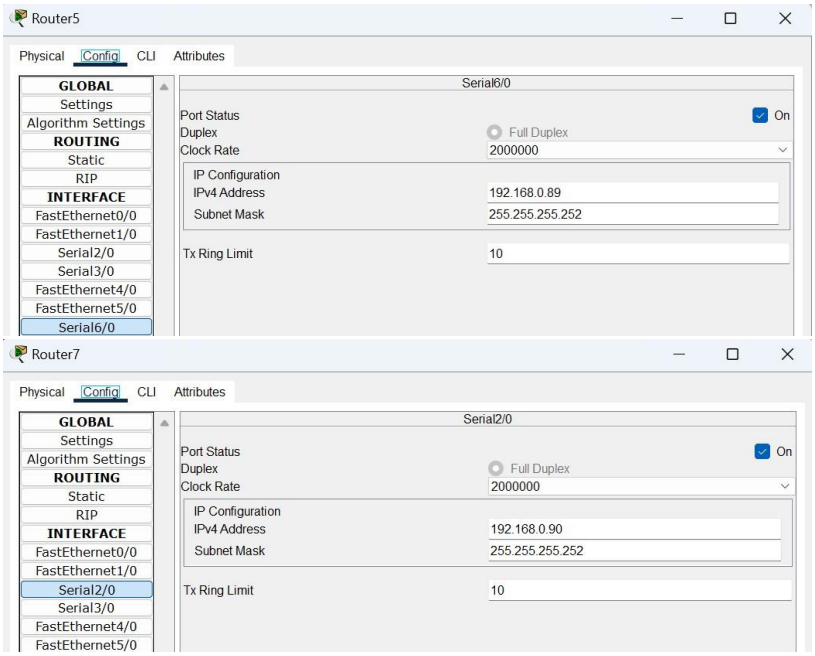
  Broadcast: 192.168.0.83



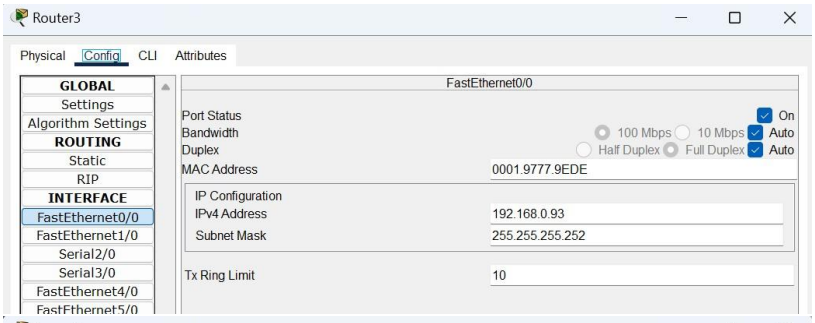
- **Subnet 10 (Router5 – Router6)  /30**
  Subnet ID: 192.168.0.84
  Host: 192.168.0.85 – 192.168.0.86
  Broadcast: 192.168.0.87

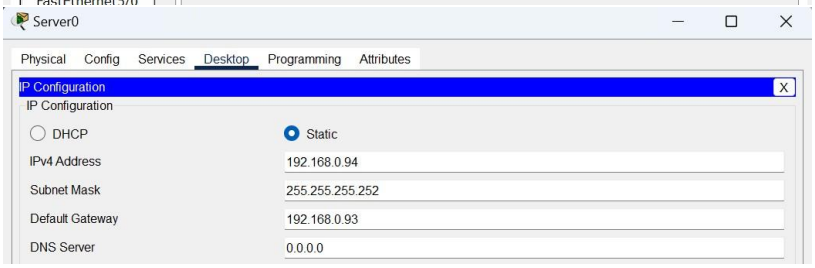



- **Subnet 11 (Router5 – Router7)  /30**
  Subnet ID: 192.168.0.88
  Host: 192.168.0.89 – 192.168.0.90
  Broadcast: 192.168.0.91




- **Subnet 12 (Router3 – Server0)  /30**
  Subnet ID: 192.168.0.92
  Host: 192.168.0.93 – 192.168.0.94
  Broadcast: 192.168.0.95



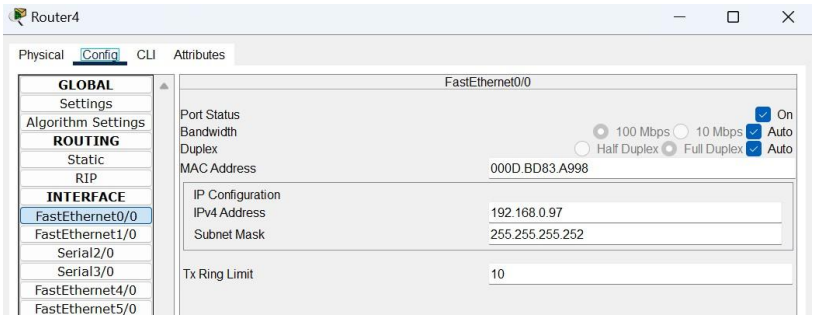



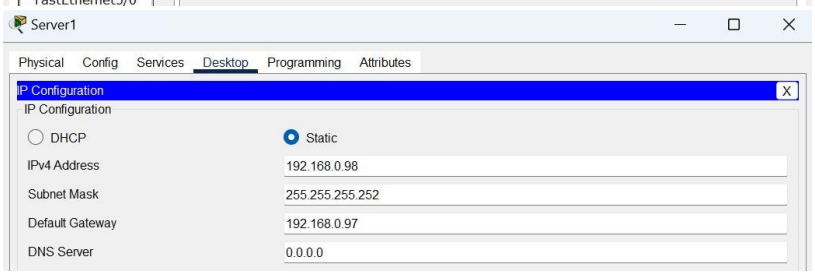

- **Subnet 13 (Router4 – Server1) /30**

  Subnet ID: 192.168.0.96

  Host: 192.168.0.97 – 192.168.0.98

  Broadcast: 192.168.0.99



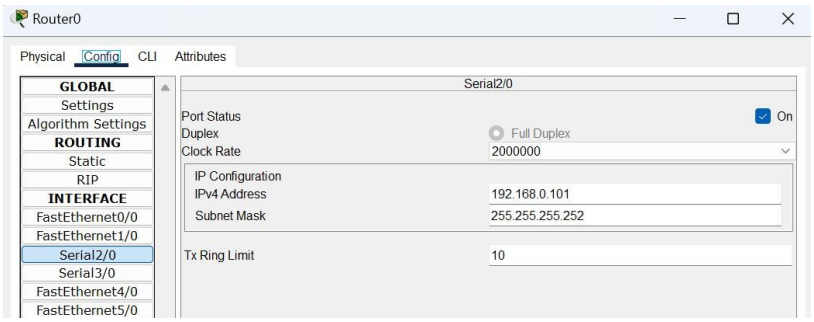



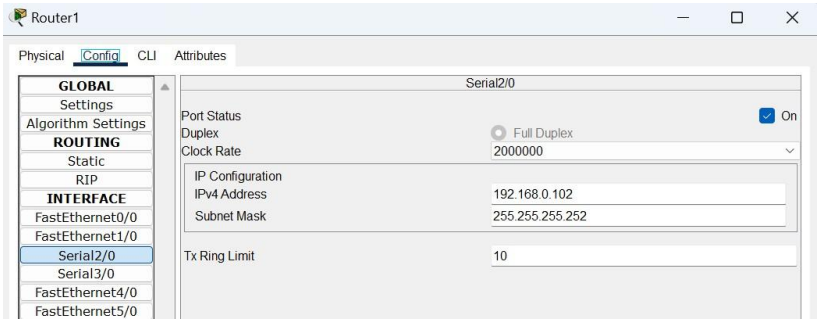

- **Subnet 14 (Router0 – Router1) /30**

  Subnet ID: 192.168.0.100

  Host: 192.168.0.101 – 192.168.0.102

  Broadcast: 192.168.0.103



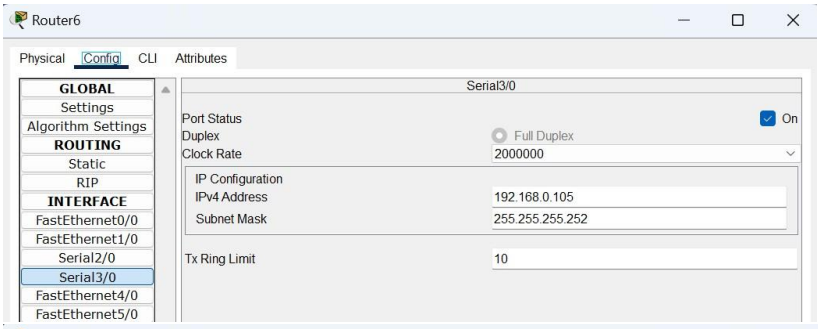


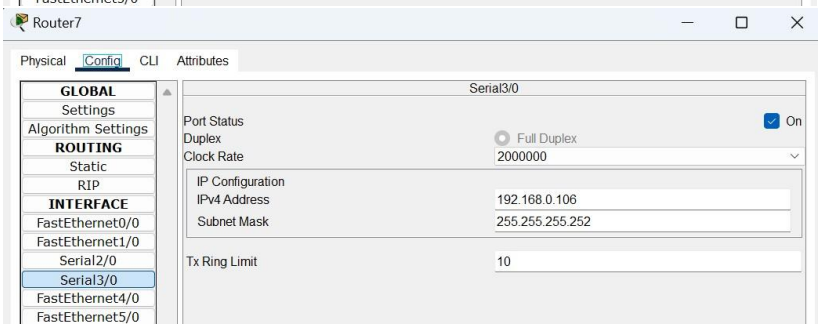

- **Subnet 15 (Router6 – Router7) /30**

  Subnet ID: 192.168.0.104

  Host: 192.168.0.105 – 192.168.0.106

  Broadcast: 192.168.0.107




## C. Konfigurasi Routing

### 1. OSPF

#### ◆ Router0

```
Router(config)#router ospf 10
Router(config-router)#network 192.168.0.100 0.0.0.3 area 0
Router(config-router)#network 192.168.0.64 0.0.0.3 area 0
Router(config-router)#network
01:00:44: %OSPF-5-ADJCHG: Process 10, Nbr 3.3.3.3 on Serial3/0 from LOADING to FULL, Loading
Done

% Incomplete command.
Router(config-router)#network 192.168.0.0 0.0.0.15 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

#### ◆ Router1

```
Router(config)#router ospf 10
Router(config-router)#network 192.168.0.100 0.0.0.3 area 0
Router(config-router)#netwo
01:02:17: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.0.101 on Serial2/0 from LOADING to FULL,
Loading Done

% Incomplete command.
Router(config-router)#network 192.168.0.68 0.0.0.3 area 0
Router(config-router)#
01:02:56: %OSPF-5-ADJCHG: Process 10, Nbr 3.3.3.3 on Serial3/0 from LOADING to FULL, Loading
Done

Router(config-router)#network 192.168.0.16 0.0.0.15 area 0
Router(config-router)#end
```

- **Router2**

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 10
Router(config-router)#network 3.3.3.3 0.0.0.0 area 0
Router(config-router)#network 192.168.0.68 0.0.0.3 area 0
Router(config-router)#network 192.168.0.64 0.0.0.3 area 0
Router(config-router)#network 192.168.0.72 0.0.0.3 area 0
Router(config-router)#
00:25:08: %OSPF-5-ADJCHG: Process 10, Nbr 1.1.1.1 on Serial6/0 from LOADING to FULL, Loading
Done
```

- **Router3**

```
Router(config)#router ospf 10
Router(config-router)#network 1.1.1.1 0.0.0.0 area 0
Router(config-router)#network 192.168.0.76 0.0.0.3 area 0
Router(config-router)#network 192.168.0.72 0.0.0.3 area 0
Router(config-router)#network 192.168.0.92 0.0.0.3 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

00:20:16: %OSPF-5-ADJCHG: Process 10, Nbr 2.2.2.2 on Serial3/0 from LOADING to FULL, Loading
Done

00:25:08: %OSPF-5-ADJCHG: Process 10, Nbr 3.3.3.3 on Serial2/0 from LOADING to FULL, Loading
Done
```

- **Router4**

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 10
Router(config-router)#network 2.2.2.2 0.0.0.0 area 0
Router(config-router)#network 192.168.0.76 0.0.0.3 area 0
Router(config-router)#network 192.168.0.76 0.0.0.3 area 0
00:20:16: %OSPF-5-ADJCHG: Process 10, Nbr 1.1.1.1 on Serial2/0 from LOADING to FULL, Loading
Done

Router(config-router)#network 192.168.0.80 0.0.0.3 area 0
Router(config-router)#network 192.168.0.96 0.0.0.3 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

00:30:37: %OSPF-5-ADJCHG: Process 10, Nbr 4.4.4.4 on Serial3/0 from LOADING to FULL, Loading
Done
```

- **Router5**

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 10
Router(config-router)#network 4.4.4.4 0.0.0.0 area 0
Router(config-router)#network 192.168.0.80 0.0.0.3 area 0
Router(config-router)#netw
00:30:37: %OSPF-5-ADJCHG: Process 10, Nbr 2.2.2.2 on Serial2/0 from LOADING to FULL, Loading
Done

% Incomplete command.
Router(config-router)#network 192.168.0.84 0.0.0.3 area 0
Router(config-router)#network 192.168.0.88 0.0.0.3 area 0
Router(config-router)#
```

- **Router6**

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 10
Router(config-router)#network 192.168.0.84 0.0.0.3 area 0
Router(config-router)#
00:51:54: %OSPF-5-ADJCHG: Process 10, Nbr 4.4.4.4 on Serial2/0 from LOADING to FULL, Loading
Done

Router(config-router)#network 192.168.0.102 0.0.0.3 area 0
Router(config-router)#network 192.168.0.32 0.0.0.15 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

- **Router7**

```
Router(config)#router ospf 10
Router(config-router)#network 192.168.0.88 0.0.0.3 area 0
Router(config-router)#network 1
00:56:19: %OSPF-5-ADJCHG: Process 10, Nbr 4.4.4.4 on Serial2/0 from LOADING to FULL, Loading
Done
                              ^
% Invalid input detected at '^' marker.

Router(config-router)#network 192.168.0.102 0.0.0.3 area 0
Router(config-router)#network 192.168.0.48 0.0.0.15 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

## 2. BGP

- **Router3**

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router bgp 100
Router(config-router)#neighbor 192.168.0.78 remote-as 101
Router(config-router)#network 192.168.0.92 mask 255.255.255.252
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
%BGP-5-ADJCHANGE: neighbor 192.168.0.78 Up
```

- **Router4**

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router bgp 101
Router(config-router)#neighbor 192.168.0.77 remote-as 100
Router(config-router)#%BGP-5-ADJCHANGE: neighbor 192.168.0.77 Up

Router(config-router)#network 192.168.0.96 mask 255.255.255.252
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

3. **Redistribusi BGP – OSPF**

- **Router3**

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router bgp 100
Router(config-router)#network 1.1.1.1 mask 255.255.255.255
Router(config-router)#redistribute ospf 10
Router(config-router)#ex
```

- **Router4**

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router bgp 101
Router(config-router)#network 2.2.2.2 mask 255.255.255.255
Router(config-router)#redistribute ospf 10
Router(config-router)#ex
```

4. **Redistribusi OSPF – BGP**

- **Router3**

```
Router(config)#router ospf 10
Router(config-router)#redistribute bgp 100 subnets
Router(config-router)#ex
```

- **Router4**

```
Router(config)#router ospf 10
Router(config-router)#redistribute bgp 101 subnets
Router(config-router)#ex
```

## D. ACL (Extended, Router5)

- **Blocking**

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#access-list 100 deny tcp host 192.168.0.34 host 192.168.0.98 eq www
Router(config)#access-list 100 deny tcp host 192.168.0.51 host 192.168.0.94 eq www
Router(config)#access-list 100 permit ip any any
Router(config)#ex
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

- **Penerapan pada interface**

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int se3/0
Router(config-if)#ip access-group 100 in
Router(config-if)#ex
Router(config)#int se6/0
Router(config-if)#ip access-group 100 in
Router(config-if)#ex
Router(config)#ex
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

## E. NAT (Overload/PAT)

1. **Router0**

- **Konfigurasi interface sisi inside**

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.0.1 255.255.255.240
Router(config-if)#ip nat inside
Router(config-if)#ex
```

- **Konfigurasi interface sisi outside**

```
Router(config)#int se3/0
Router(config-if)#ip address 192.168.0.65 255.255.255.252
Router(config-if)#ip nat outside
Router(config-if)#ex
```

- **Konfigurasi ACL**

```
Router(config)#
Router(config)#access-list 1 permit 192.168.0.0 0.0.0.15
```

- **Mengaktifkan NAT overload (PAT)**

```
Router(config)#ip nat inside source list 1 interface se3/0 overload
Router(config)#
```

2. **Router1**

- **Konfigurasi interface sisi inside**

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.0.17 255.255.255.240
Router(config-if)#ip nat inside
Router(config-if)#ex
```

- **Konfigurasi interface sisi outside**

```
Router(config)#int se3/0
Router(config-if)#ip address 192.168.0.69 255.255.255.252
Router(config-if)#ip nat outside
Router(config-if)#ex
```

- **Konfigurasi ACL**

```
Router(config)#
Router(config)#access-list 1 permit 192.168.0.16 0.0.0.15
```

- **Mengaktifkan NAT overload (PAT)**

```
Router(config)#ip nat inside source list 1 interface se3/0 overload
Router(config)#
```

## F. Hasil Pengujian dan Analisa

### 1. OSPF (Router2)

- **Router#show ip ospf neighbor**

```
Router#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.0.102     0   FULL/  -        00:00:35    192.168.0.69    Serial2/0
192.168.0.101     0   FULL/  -        00:00:35    192.168.0.65    Serial3/0
192.168.0.93      0   FULL/  -        00:00:31    192.168.0.74    Serial6/0
Router#
```

- **Router#show ip route**

```
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.0.0/24 is variably subnetted, 15 subnets, 2 masks
O       192.168.0.0/28 [110/65] via 192.168.0.65, 01:25:23, Serial3/0
O       192.168.0.16/28 [110/65] via 192.168.0.69, 01:25:23, Serial2/0
O       192.168.0.32/28 [110/257] via 192.168.0.74, 00:54:33, Serial6/0
O       192.168.0.48/28 [110/257] via 192.168.0.74, 01:25:23, Serial6/0
C       192.168.0.64/30 is directly connected, Serial3/0
C       192.168.0.68/30 is directly connected, Serial2/0
C       192.168.0.72/30 is directly connected, Serial6/0
O       192.168.0.76/30 [110/128] via 192.168.0.74, 01:25:23, Serial6/0
O       192.168.0.80/30 [110/192] via 192.168.0.74, 01:25:23, Serial6/0
O       192.168.0.84/30 [110/256] via 192.168.0.74, 00:54:33, Serial6/0
O       192.168.0.88/30 [110/256] via 192.168.0.74, 01:25:23, Serial6/0
O       192.168.0.92/30 [110/65] via 192.168.0.74, 01:25:23, Serial6/0
O       192.168.0.96/30 [110/129] via 192.168.0.74, 01:25:23, Serial6/0
O       192.168.0.100/30 [110/128] via 192.168.0.65, 01:25:23, Serial3/0
                         [110/128] via 192.168.0.69, 01:25:23, Serial2/0
O       192.168.0.104/30 [110/320] via 192.168.0.74, 01:25:23, Serial6/0
```

- **Analisa**

Command pertama digunakan untuk menampilkan hasil pengujian router yang memiliki relasi neighbor. Di mana pada hasil pengujian router2 tersebut, diketahui bahwa router neighbor dari router2 adalah router0 (192.168.0.65) yang terhubung melalui interface se2/0 milik router2, router1 (192.168.0.69) terhubung melalui interface se3/0 milik router2, serta router3 (192.168.0.74) yang terhubung melalui interface se6/0 milik router2.

Command kedua adalah hasil pengujian untuk mengetahui routing table dari suatu router. Di mana dari hasil tersebut diketahui bahwa router2 terkoneksi secara langsung pada 3 subnet, yaitu subnet dari router-router yang terhubung langsung dengan router2, meliputi router0 (network address 192.168.0.64), router1 (network address 192.168.0.68) dan router3 (network address 192.168.0.72). Selain itu, untuk ke 12 subnet lainnya, terhubung secara OSPF (O).

## 2. BGP (Router3)

- **Router#show ip bgp**

```
Router#show ip bgp
BGP table version is 5, local router ID is 192.168.0.93
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*> 192.168.0.92/30   0.0.0.0                  0      0 32768 i
*> 192.168.0.96/30   192.168.0.78             0      0      0 101 i
```

- **Router#show ip route**

```
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.0.0/24 is variably subnetted, 15 subnets, 2 masks
O       192.168.0.0/28 [110/129] via 192.168.0.73, 01:27:44, Serial2/0
O       192.168.0.16/28 [110/129] via 192.168.0.73, 01:27:44, Serial2/0
O       192.168.0.32/28 [110/193] via 192.168.0.78, 00:56:59, Serial3/0
O       192.168.0.48/28 [110/193] via 192.168.0.78, 01:27:44, Serial3/0
O       192.168.0.64/30 [110/128] via 192.168.0.73, 01:27:44, Serial2/0
O       192.168.0.68/30 [110/128] via 192.168.0.73, 01:27:44, Serial2/0
C       192.168.0.72/30 is directly connected, Serial2/0
C       192.168.0.76/30 is directly connected, Serial3/0
O       192.168.0.80/30 [110/128] via 192.168.0.78, 01:27:44, Serial3/0
O       192.168.0.84/30 [110/192] via 192.168.0.78, 00:56:59, Serial3/0
O       192.168.0.88/30 [110/192] via 192.168.0.78, 01:27:44, Serial3/0
C       192.168.0.92/30 is directly connected, FastEthernet0/0
B       192.168.0.96/30 [20/0] via 192.168.0.78, 00:00:00
O       192.168.0.100/30 [110/192] via 192.168.0.73, 01:27:44, Serial2/0
O       192.168.0.104/30 [110/256] via 192.168.0.78, 01:27:44, Serial3/0
```

- **Analisa**

Command pertama digunakan untuk menampilkan hasil pengujian mengenai informasi dari routing BGP yang telah dilakukan, di mana akan muncul ketika BGP berhasil diterapkan. BGP pada router3, diterapkan untuk router4 (router neighbor) yang sudah berbeda AS serta server0. Tabel kolom network pada hasil pengujian menunjukkan network-network yang terhubung dengan router3 dengan BGP, meliputi network 192.168.0.92 (server0) yang terindikasi terhubung secara langsung dengan router3 (Next Hop = 0.0.0.0) dan network 192.168.0.96 (server1)

yang terindikasi terhubung tidak langsung dengan router3 dan harus melalui router4 terlebih dahulu (Next Hop = 192.168.0.78).

Command kedua digunakan untuk menampilkan table routing dari router3. Di mana dari hasil tersebut diketahui bahwa router3 terkoneksi secara langsung (C) dengan router2 (network address 192.168.0.72), router4 (network address 192.168.0.76), dan server0 (network address 192.168.0.92). Dan terkoneksi secara BGP (B) dengan server1 (network address 192.168.0.96). Selain itu, untuk 11 subnet sisanya terhubung secara OSPF. Hal ini disebabkan karena baik pada router3 maupun router4 dilakukan 2 konfigurasi routing dinamis, yaitu OSPF dan BGP dan juga dilakukan 2 redistribusi (BGP – OSPF dan OSPF – BGP).

## 3. ACL

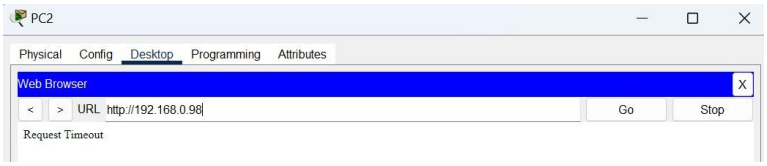
- **Router#show access-list**

```
Router>en
Router#show access-list
Standard IP access list 1
    10 permit 192.168.0.0 0.0.0.255
Extended IP access list 100
    10 deny tcp host 192.168.0.34 host 192.168.0.98 eq www
    20 deny tcp host 192.168.0.51 host 192.168.0.94 eq www
    30 permit ip any any (1924 match(es))
```

- **Pengujian konektivitas**

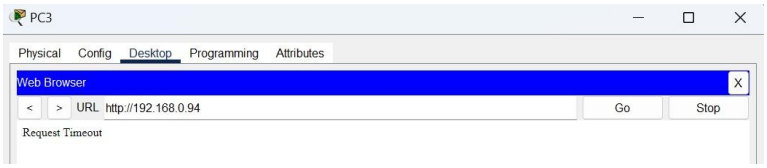
**PC2**

- Web



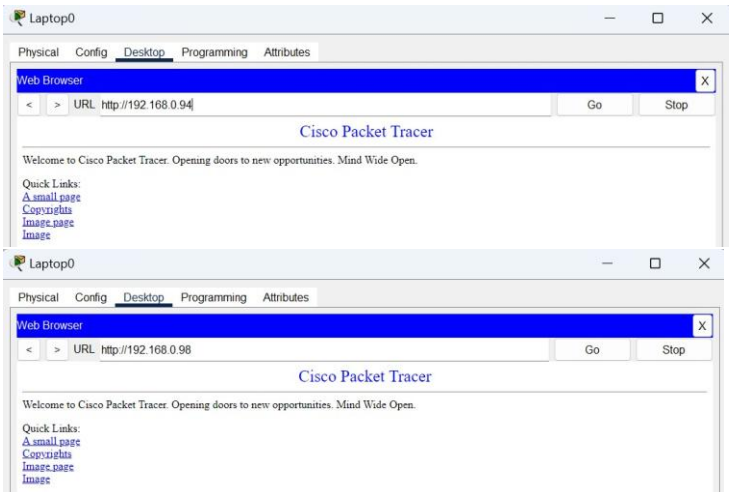

**PC3**

- Web




**Laptop0**

- Web



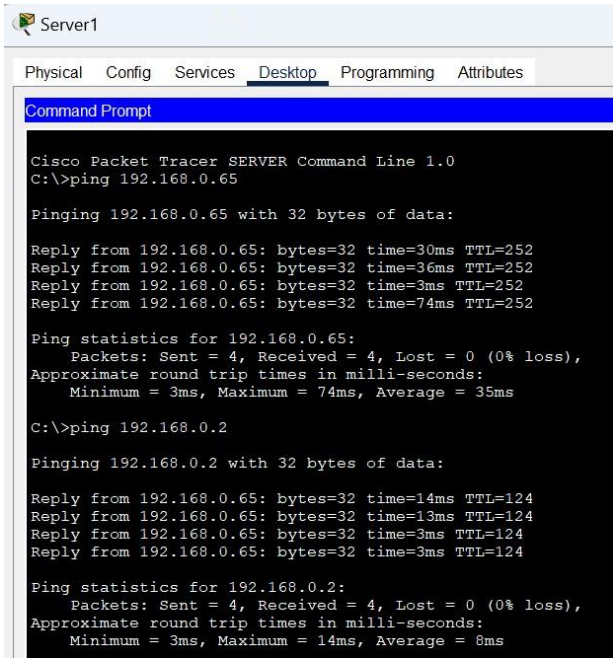
- **Analisa**

  Command tersebut digunakan untuk mengecek hasil konfigurasi ACL yang telah diterapkan pada topologi. Hasil pengujian menampilkan 2 jenis ACL, yaitu standard ACL dan extended ACL. Standard ACL ditampilkan karena telah dilakukan konfigurasi pada saat melakukan NAT overload. Di mana standard ACL tersebut merupakan bagian dari langkah-langkah dalam konfigurasi Nat overload, yang berfungsi untuk mencocokkan paket yang masuk ke inside interface baik milik router0 maupun router1.

  Sedangkan extended ACL, merupakan ACL tambahan yang diterapkan pada router5. Di mana digunakan untuk memblokir akses dari PC2 (192.168.0.34) ke web browser server1 (192.168.0.98) dan memblokir akses dari PC3 (192.168.0.51) ke web browser server0 (192.168.0.94). Sedangkan untuk akses lainnya diizinkan sesuai dengan yang ditampilkan pada hasil pengujian konektivitas. Extended ACL dipilih karena filter yang diinginkan hanya untuk beberapa jenis akses konektivitas dan untuk host tertentu. Nomor 10, 20, dan 30 menunjukkan urutan prioritas aturan yang akan dilakukan/dicek.

## 4. NAT

- **Pengujian konektivitas (Server1 – PC0)**




- **Router#show ip nat translation**

```
Router>en
Router#show ip nat translation
Pro  Inside global     Inside local       Outside local      Outside global
icmp 192.168.0.65:4    192.168.0.2:4      192.168.0.51:4     192.168.0.51:4
icmp 192.168.0.65:5    192.168.0.2:5      192.168.0.50:5     192.168.0.50:5
tcp 192.168.0.65:1026  192.168.0.2:1026   192.168.0.94:80    192.168.0.94:80
tcp 192.168.0.65:1027  192.168.0.2:1027   192.168.0.98:80    192.168.0.98:80
tcp 192.168.0.65:1028  192.168.0.2:1028   192.168.0.98:80    192.168.0.98:80
tcp 192.168.0.65:1029  192.168.0.2:1029   192.168.0.98:80    192.168.0.98:80
```

- **Analisa**

  Hasil pengujian konektivitas menunjukkan kesalahan, di mana server1 yang berada di network eksternal dapat terkoneksi dengan PC0 yang berada di network internal dengan melalui IP privat (inside local) milik PC0. Hal tersebut tidak seharusnya terjadi, karena telah diaplikasikan NAT pada router0. Di mana NAT berfungsi untuk mentranslasikan IP privat (inside local) milik device yang berada di network internal menjadi IP publik (inside global), sehingga device yang berada pada network internal dapat berkomunikasi dengan network eksternal (pada network eksternal, digunakan IP publik. Walaupun reply yang ditunjukkan tetap berasal dari IP publiknya.

  Hal tersebut mungkin saja disebabkan karena routing protokol yang digunakan (BGP dan OSPF). Di mana salah satu dari protokol tersebut mungkin saja menyebarkan rute network internal ke network eksternal, sehingga membuat alamat intenal dapat dijangkau secara eksternal. Penyebaran tersebut juga dapat disebabkan karena filter yang tidak sesuai pada redistribusi. Di mana, seperti yang diketahui, tidak dilakukan pemfilteran pada konfigurasi redistribusi di topologi ini.

  Selain itu, setelah dilakukan trafik pada topologi, informasi translasi dari pengaplikasian NAT dapat diketahui dengan menggunakan command pada gambar hasil di atas. Di mana NAT overload diterapkan pada router0 untuk mentranslasikan IP privat device yang berada pada network internal (PC0). Dan didapatkan perbedaan alamat pada inside global dan inside localnya. Di mana alamat inside local merupakan alamat privat milik PC0 yang telah ditetapkan sebelumnya, yaitu 192.168.0.2. Sedangkan inside global merupakan alamat publik yang diperoleh dari interface Serial3/0 milik router0 yang telah dikonfigurasikan sebelumnya sebagai sisi outside. Di mana IP dari PC0 yang sebelumnya 192.168.0.2 (inside local) ditranslasikan menjadi 192.168.0.65 (inside global) untuk berkomunikasi dengan network eksternal. Di mana pada NAT overload, untuk setiap entry trafik yang dilakukan tetap menggunakan IP publik (inside global) yang sama, yaitu 192.168.0.65. Pembedanya (letak keunikan address) terletak pada port yang berada dibelakang IP global, di mana untuk setiap entry trafik portnya berbeda-beda.